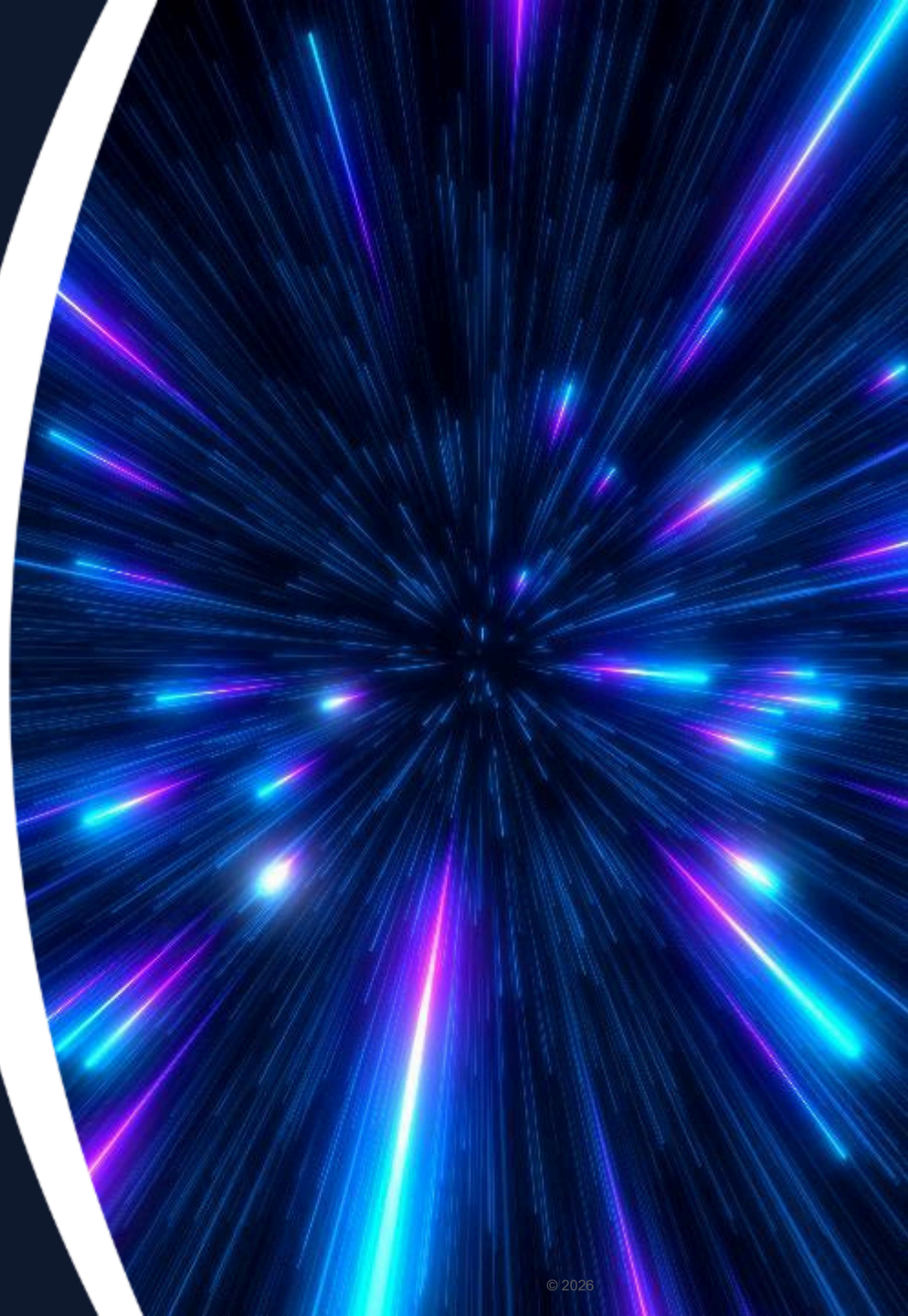
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Layer 2 Networking MAC, ARP, and VLAN

Network QA Trainees



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Contents

1. MAC
   - 1.1. MAC Denifition
   - 1.2. What is the MAC address?
   - 1.3. MAC Types
   - 1.4. MAC Address Table
   - 1.5. MAC Learning - Forwarding - Moving
   - 1.6. MAC Aging
2. ARP
   - 2.1. ARP Definition
   - 2.2. ARP Packet Structure
   - 2.3. ARP Table
   - 2.4. ARP Variants
3. VLAN
   - 3.1. VLAN Definitions
   - 3.2. Why we need VLAN
   - 3.3. Applications of VLAN
   - 3.4. 802.1Q tagged frame
   - 3.5. VLAN Port Modes
   - 3.6. VLAN Frame Handling
   - 3.7. VLAN Assosiation rules

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MAC DENIFITION

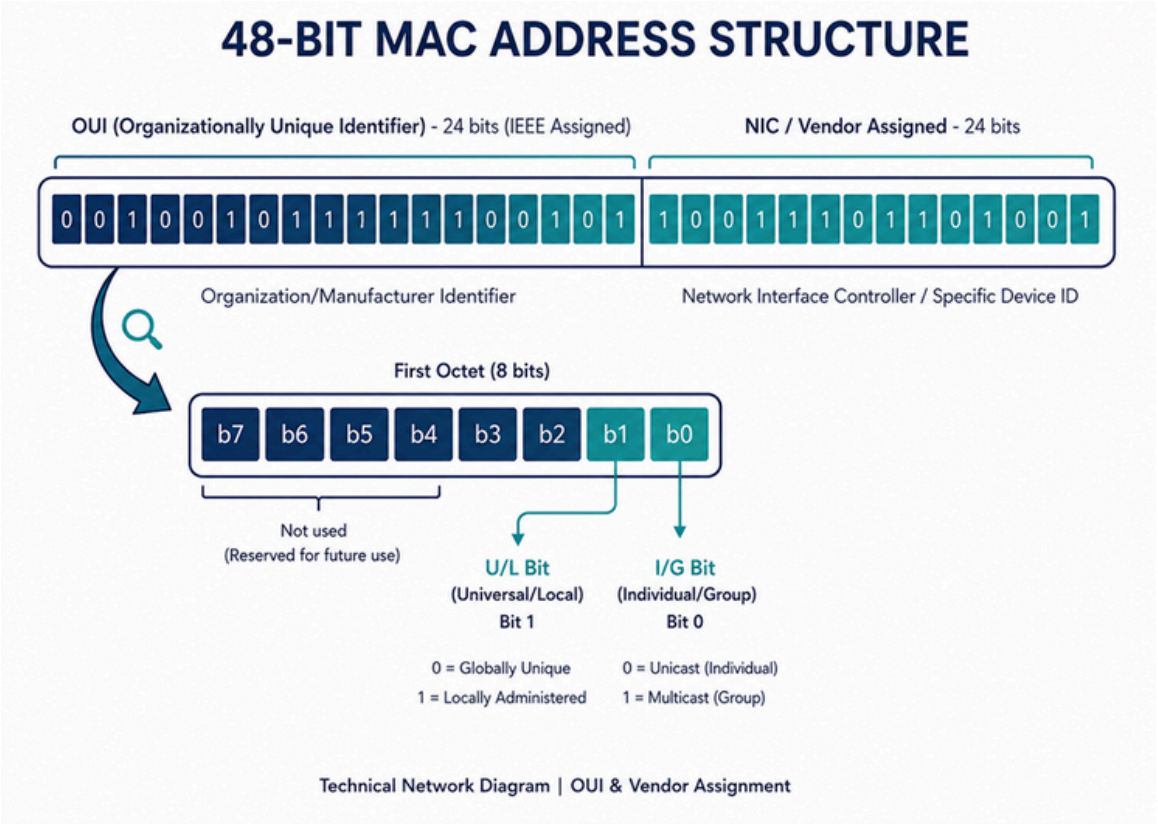
Media Access Control (MAC) is a network sublayer protocol that governs how devices share a physical communication channel to transmit data safely and efficiently.

Core Functions of MAC

- Channel Access: Determines when and how devices can send data over a shared medium.
- Collision Prevention: Stops data packets from crashing into each other during simultaneous transmission.
- Unique Addressing: Assigns and utilizes a unique hardware identifier for physical devices.
- Frame Encapsulation: Packages raw data into structured frames and appends error-checking data.

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What is a MAC Address?



A MAC address is a 48-bit (6-byte) binary string, written as 12 hex characters in pairs, separated by hyphens, colons, or periods.

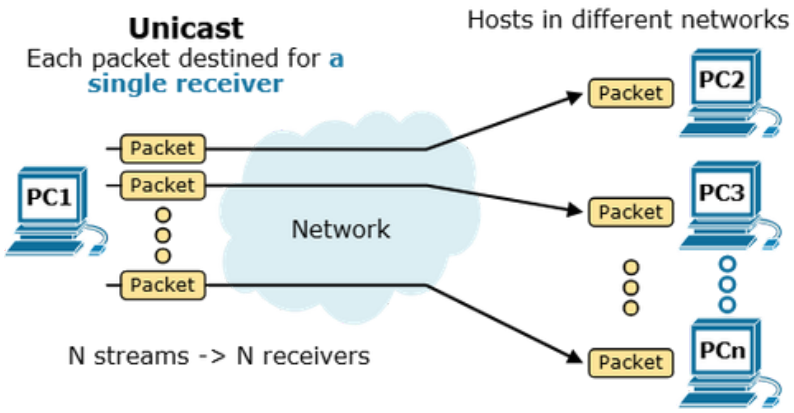
OUI: Organizationally Unique Identifier; first 24 bits. The vendor ID issued by IEEE.

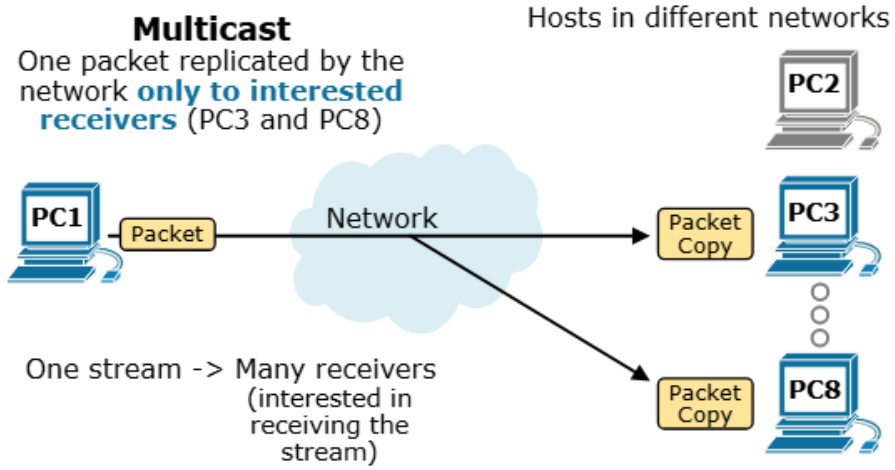
NIC-Assigned: Vendor-Assigned last 24 bits. Set by the vendor for each physical card; one OUI supports up to 2^24 devices.

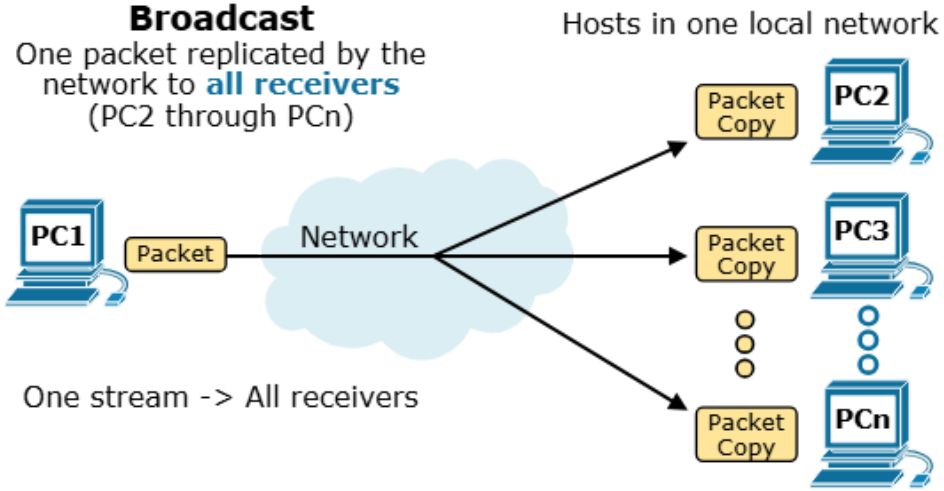
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MAC TYPES

Three types of MAC address







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MAC ADDRESS TABLE

MAC Address Table is a data table stored in the switch's CAM (Content Addressable Memory).
The table contains mappings between:

- Device MAC address
- Corresponding VLAN
- Connected port
- Entry type (dynamic or static)

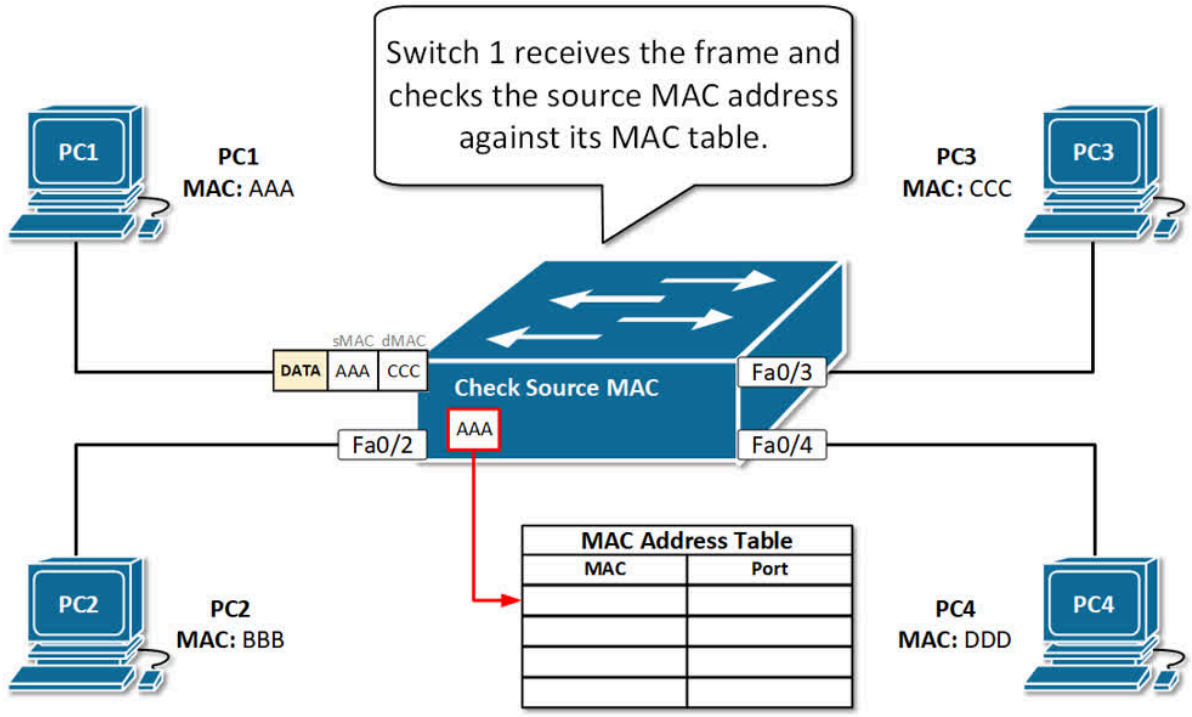
```
SwitchX#show mac address-table
          Mac Address Table
-------------------------------------------
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
 All    0008.a445.9b40    STATIC      CPU
 All    0100.0ccc.cccc    STATIC      CPU
 All    0100.0ccc.cccd    STATIC      CPU
 All    0100.0cdd.dddd    STATIC      CPU
   1    0008.e3e8.0440    DYNAMIC     Fa0/2
Total Mac Addresses for this criterion: 5
SwitchX#
```

This photo is for illustrative purposes only.Source: Internet

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MAC LEARNING

A switch learns the source MAC address of every incoming frame before forwarding it

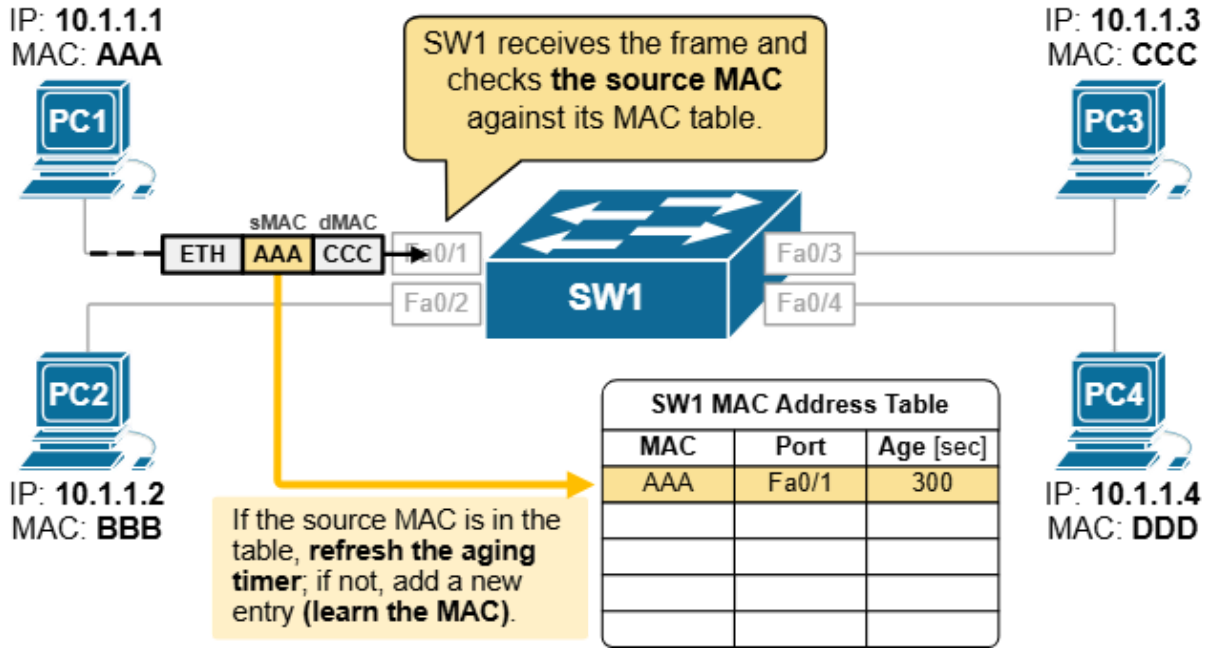


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MAC FORWARDING

After learning the Source MAC Address, the switch checks the Destination MAC Address to determine how to forward the frame.

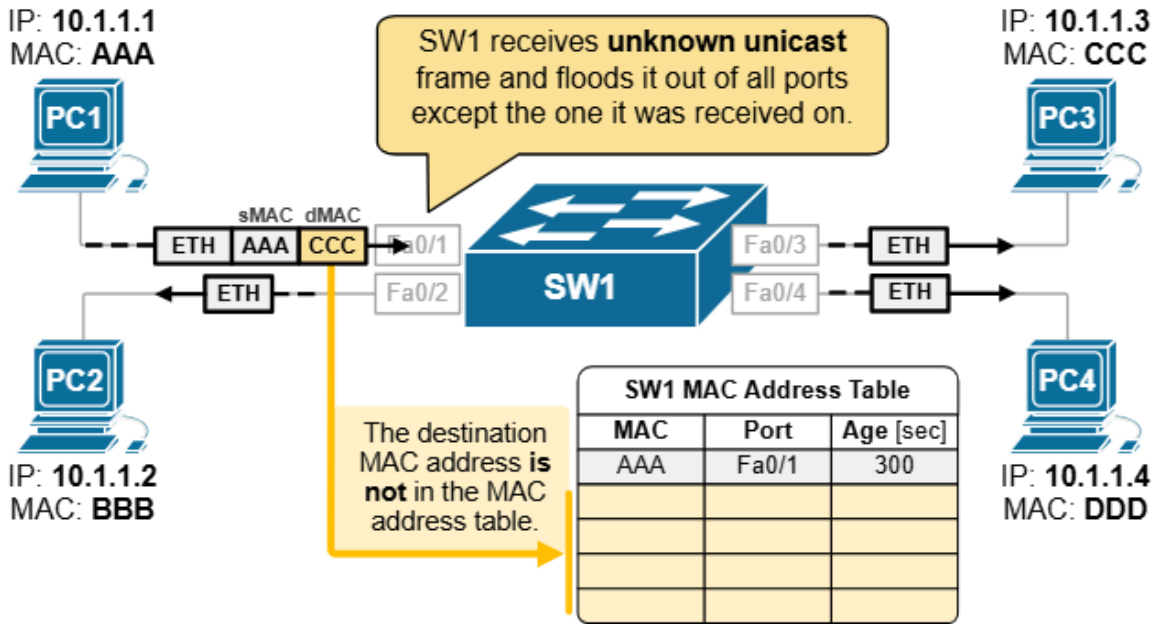
- If the destination MAC address is already in the MAC Address Table, the frame is forwarded directly to the corresponding port.

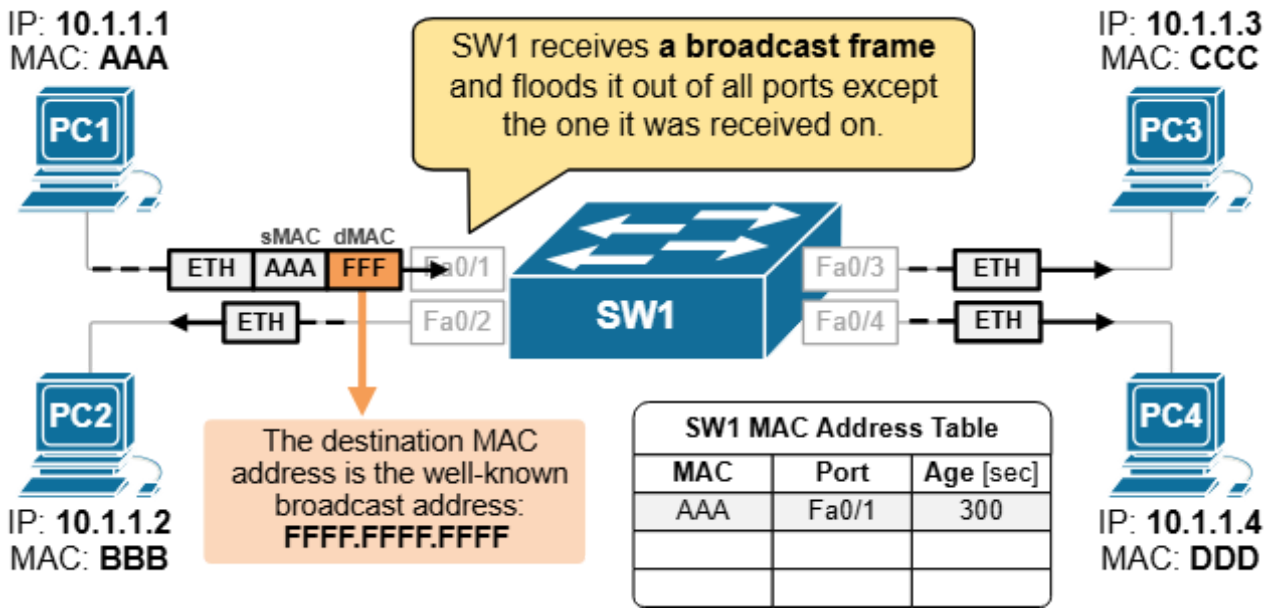


This photo is for illustrative purposes only. Source: Internet

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MAC FORWARDING



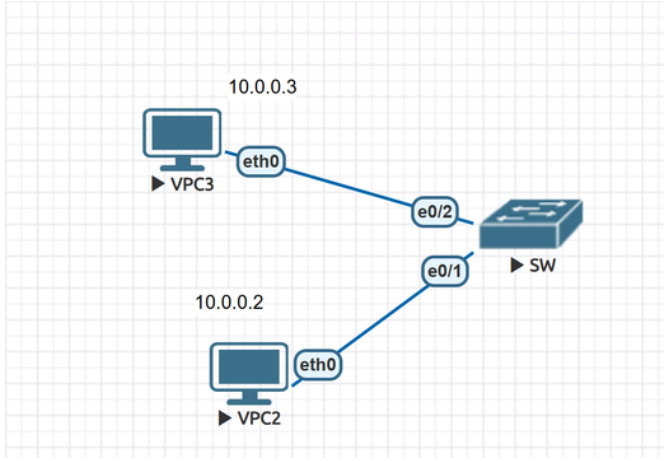


- If the destination MAC address is unknown, the switch floods the frame to all other ports in the same VLAN, except the incoming port.
- If the destination address is a broadcast or multicast, the switch also floods the frame out of all ports except on the port the frame came in.

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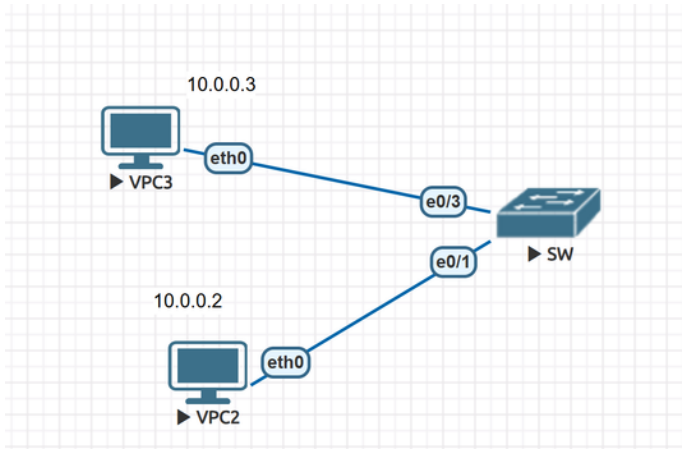
MAC MOVING

A MAC address is considered to have been moved when the same MAC address is received on a different interface in the same VLAN.




```
Switch#show mac add
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
   1    0050.7966.6802    DYNAMIC     Et0/1
   1    0050.7966.6803    DYNAMIC     Et0/2
Total Mac Addresses for this criterion: 2
```




```
Switch#show mac add
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
   1    0050.7966.6802    DYNAMIC     Et0/1
   1    0050.7966.6803    DYNAMIC     Et0/3
Total Mac Addresses for this criterion: 2
```

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MAC AGING

WHY MAC AGING IS CRUCIAL

CAM Resource Management

Content Addressable Memory (CAM) has finite hardware capacity. Aging purges stale entries from disconnected or inactive devices, reclaiming critical memory space.

Topology Adaptability

When devices move across switch ports or failover occurs, obsolete MAC-to-port mappings are flushed so traffic can immediately adapt to the updated path.

Standardized Timeout Balance

The IEEE 802.1D specification defines 300 seconds as the industry standard timeout.

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MAC AGING

LIFECYCLE OF A MAC ENTRY

1. Learning

Frame arrives. Switch binds Source MAC to Port/VLAN and starts Timer.

2. Refresh

Subsequent active frames from same MAC reset Aging Timer.

3. Purge

Host goes silent. Timer reaches timeout threshold and flushes MAC.

4. Flooding

New inbound frame flooded as Unknown Unicast until relearned.

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MAC AGING

LIFECYCLE OF A MAC ENTRY

SWITCH MAC / CAM TABLE AGING

TIMER REFRESH

VLAN	MAC ADDRESS	PORT	AGING TIMER	STATUS
10	00:1a:2b:3c:4d:5e	Gi0/1	300s	Refreshed

2. New frame received from Device A -> Switch resets Aging Timer back to 300s (Hit Bit = 1).

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ARP- Address Resolution Protocol

Logical Address (Layer 3)

- **Protocol-Based:** addresses; dynamic and changeable.
- **Hierarchical:** Split into Network ID & Host ID.
- **Routing Purpose:** Enables inter-network path finding.

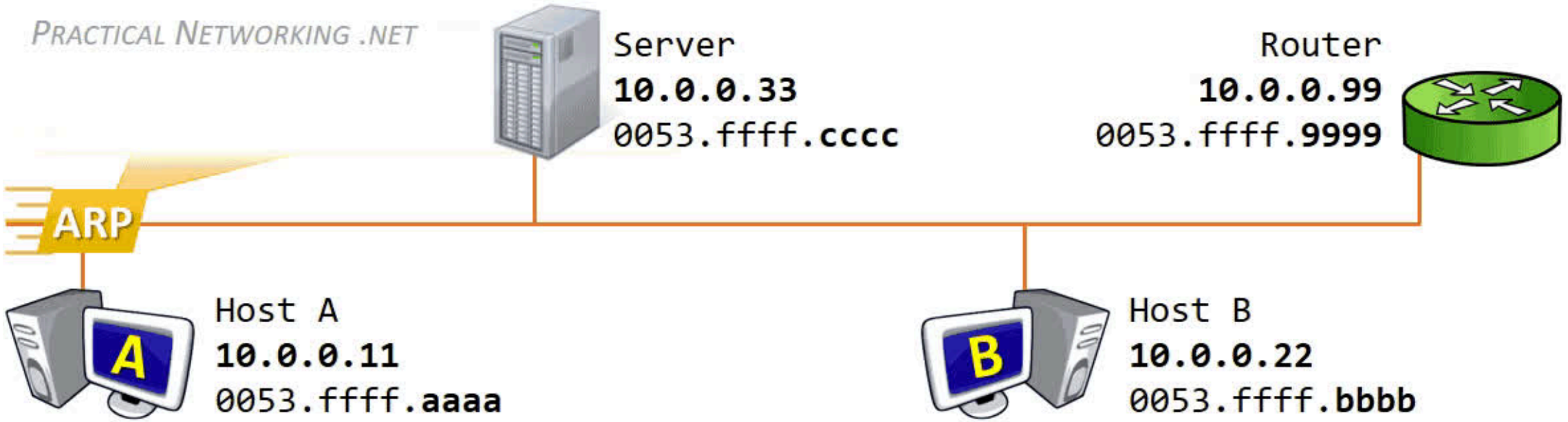
Physical Address (Layer 2)

- **Hardware-Based:** 48-bit address burned into the NIC.
- **Flat Structure:** Unique per device; non-routable globally.
- **Delivery Purpose:** Enables direct link-level frame delivery.

Because NICs only communicate using MAC addresses at Layer 2, **ARP** dynamically maps logical IP addresses to physical MAC addresses for local transmission.

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ARP DEFINITION



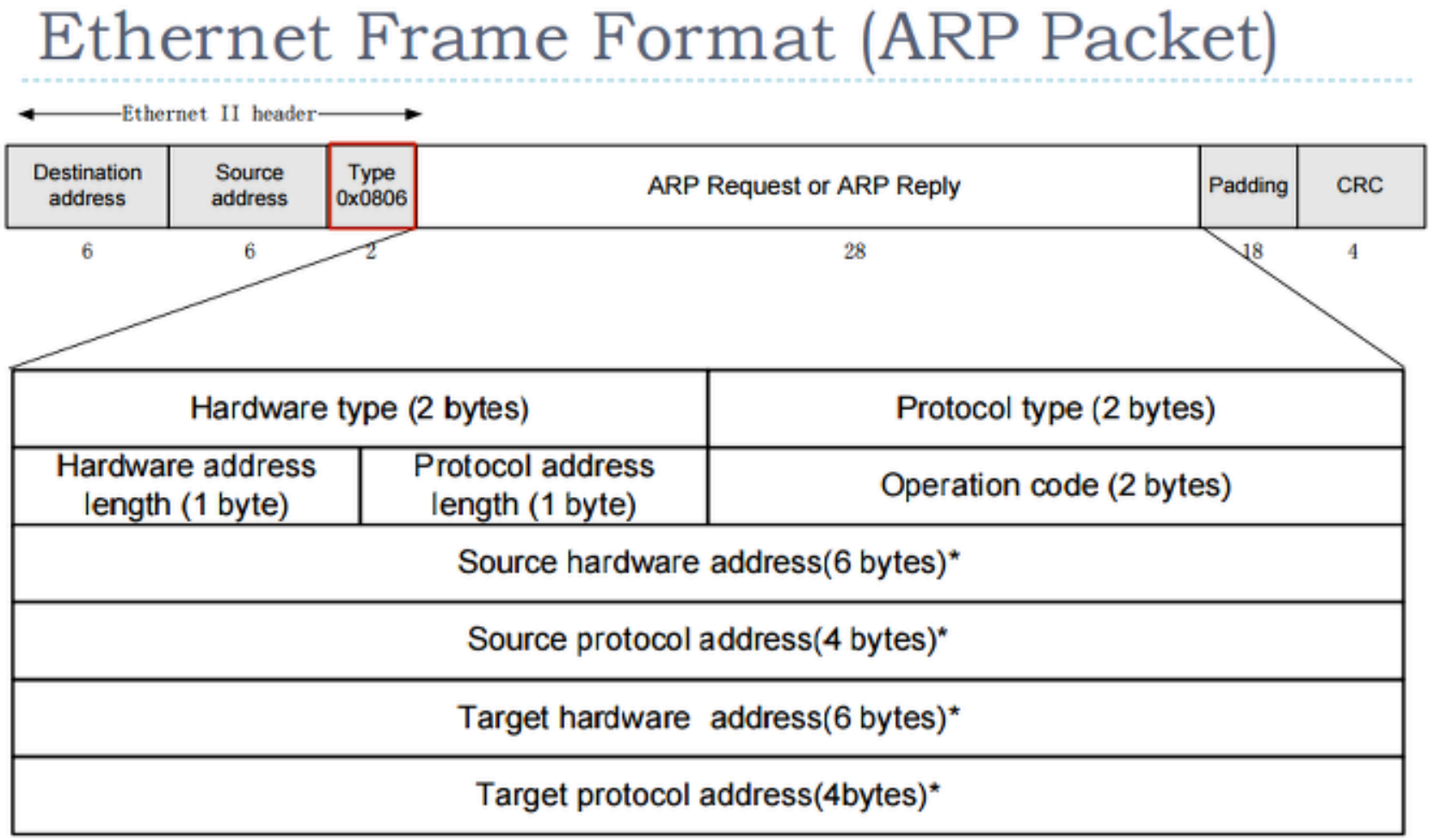
Primary Purpose: Resolves destination MAC addresses to construct complete Layer 2 Ethernet Frames.

Operational Scope: Operates strictly within the local broadcast domain / VLAN.

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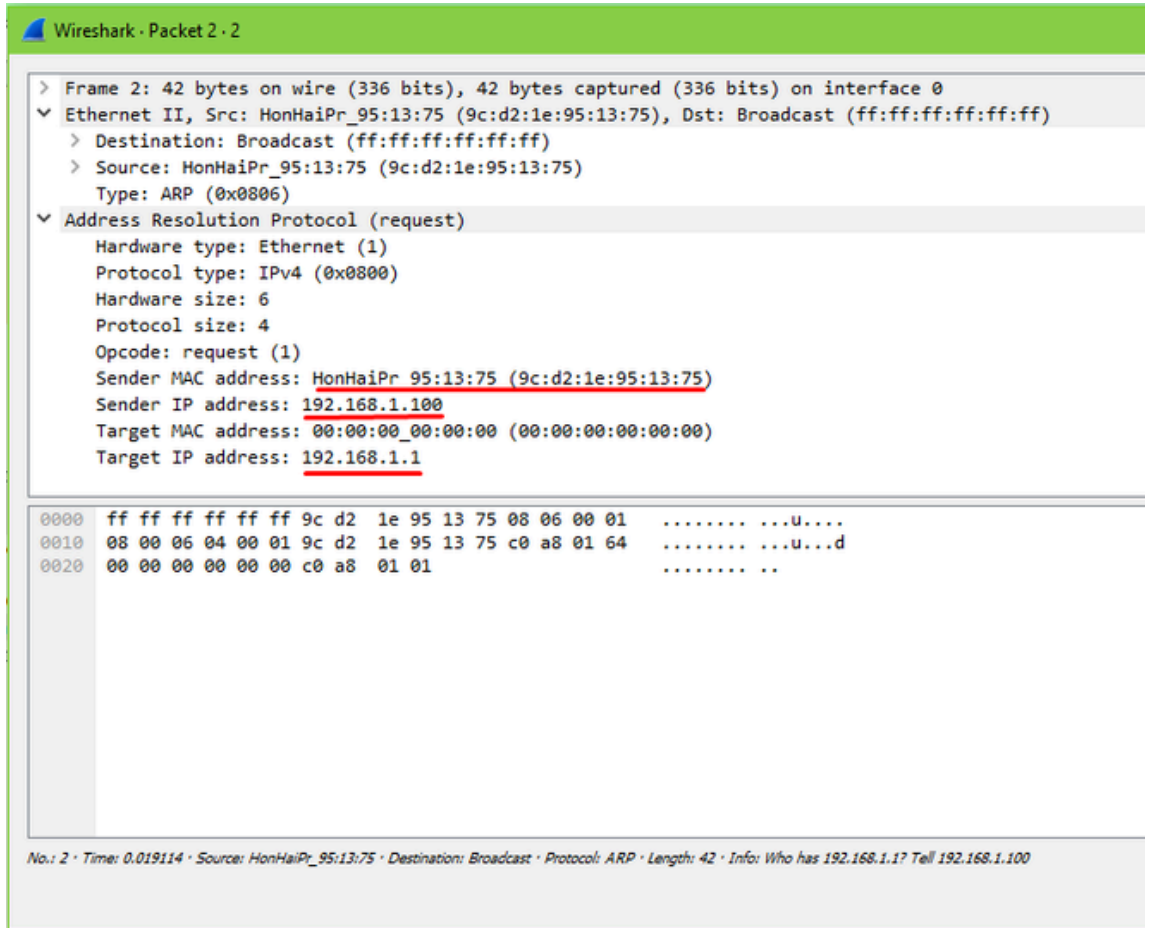
ARP PACKET STRUCTURE

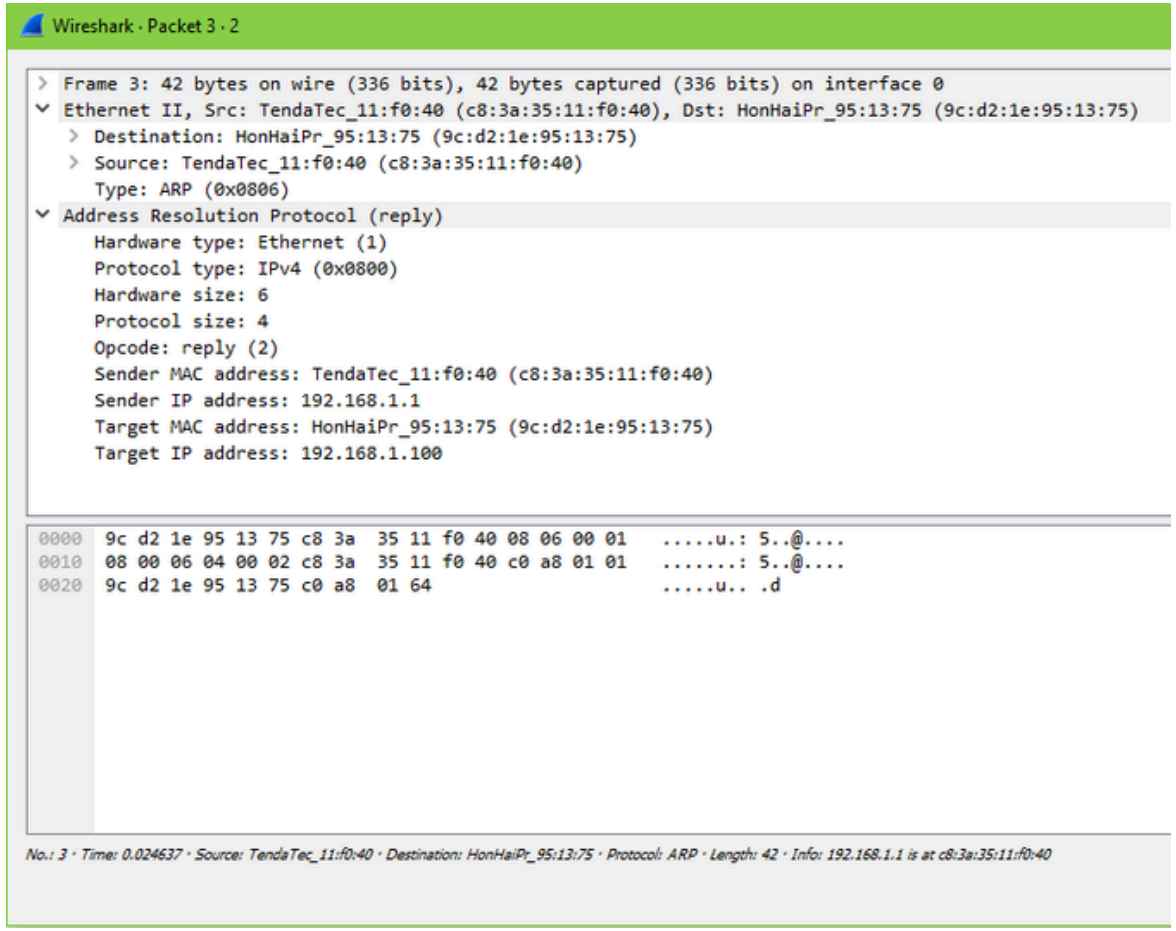
Standard IPv4 over Ethernet ARP packets carry a fixed lenghth of 28 bytes



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ARP PACKET STRUCTURE





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ARP TABLE

- The ARP Table stores IP-to-MAC mappings on a device
  → Reduce the number of ARP requests
- ARP entries are learned from received ARP messages

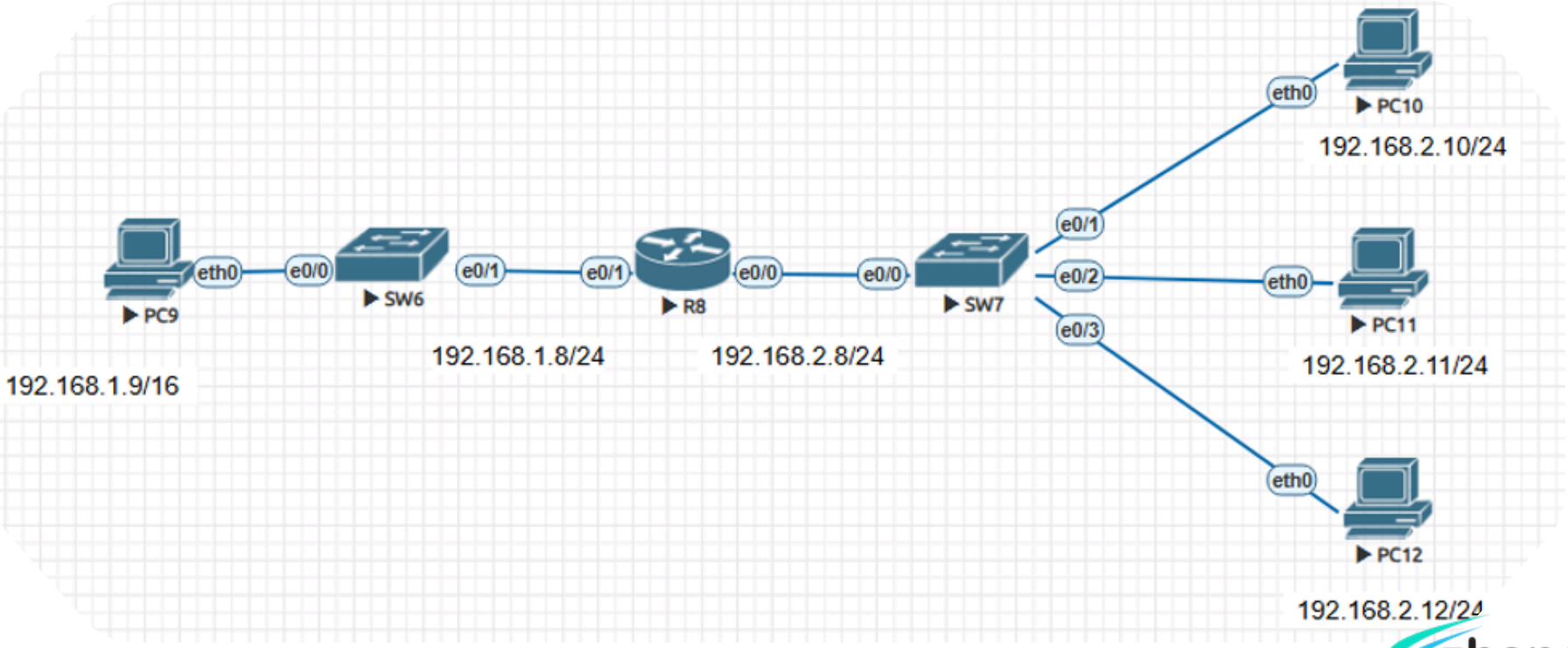
```
Router#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.8            -   aabb.cc00.8010  ARPA   Ethernet0/1
Internet  192.168.1.9            4   0050.7966.6809  ARPA   Ethernet0/1
Internet  192.168.2.8            -   aabb.cc00.8000  ARPA   Ethernet0/0
Internet  192.168.2.10           4   0050.7966.680a  ARPA   Ethernet0/0
```

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ARP VARIANTS

1. Proxy ARP

- An intermediate device responds to an ARP request on behalf of a remote host and provides its own MAC address.
- ARP proxy helps hosts that are unaware of their subnet boundaries or have improperly configured subnets.



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ARP VARIANTS

2. ARP Probe

Used to check whether an IPv4 address is already in use on the local network before a device starts using that address.

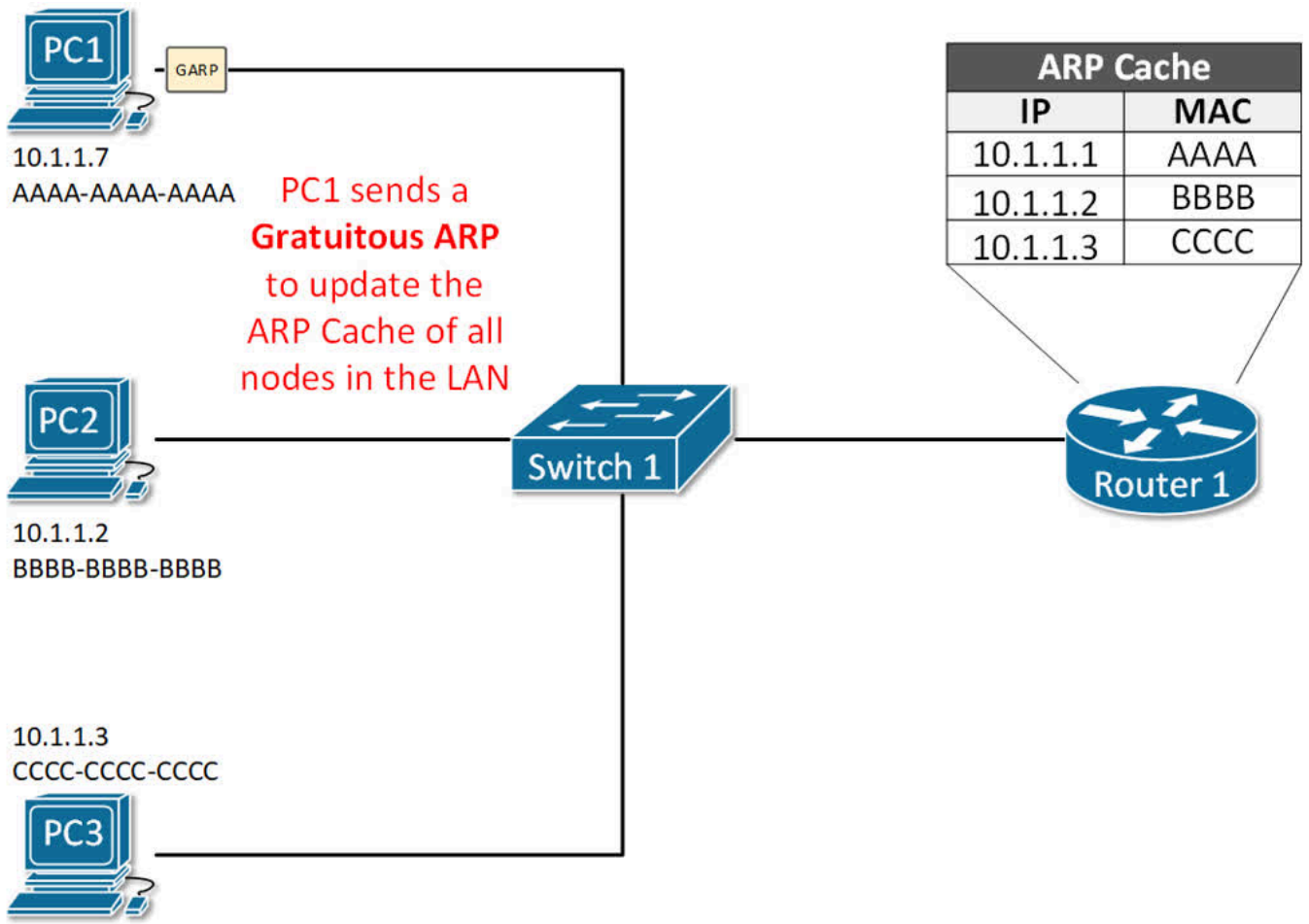
Field	Value
SHA (Sender Hardware Address)	MAC address of the device performing the probe
SPA (Sender Protocol Address)	0.0.0.0
THA (Target Hardware Address)	00:00:00:00:00:00:00:00 (recommendation by RFC 5227)
TPA (Target Protocol Address)	The IPv4 address being checked

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ARP VARIANTS

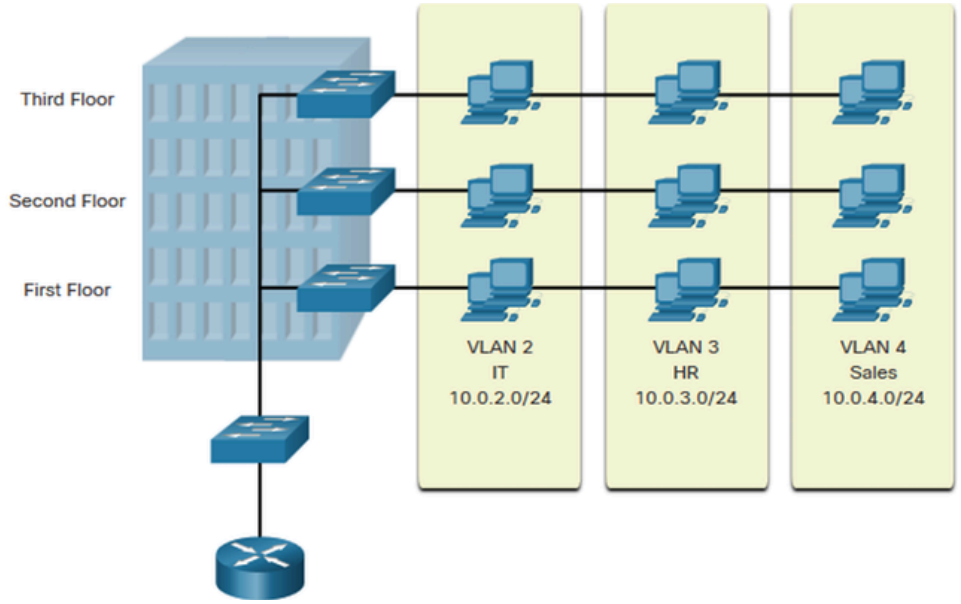
3. Gratuitous ARP (GARP)

- Mainly used to notify other hosts in the network of a change in MAC or IP address.
- Update the sender's IP–MAC mapping.
- Sender IP Address = Target IP Address



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VLAN Definitions

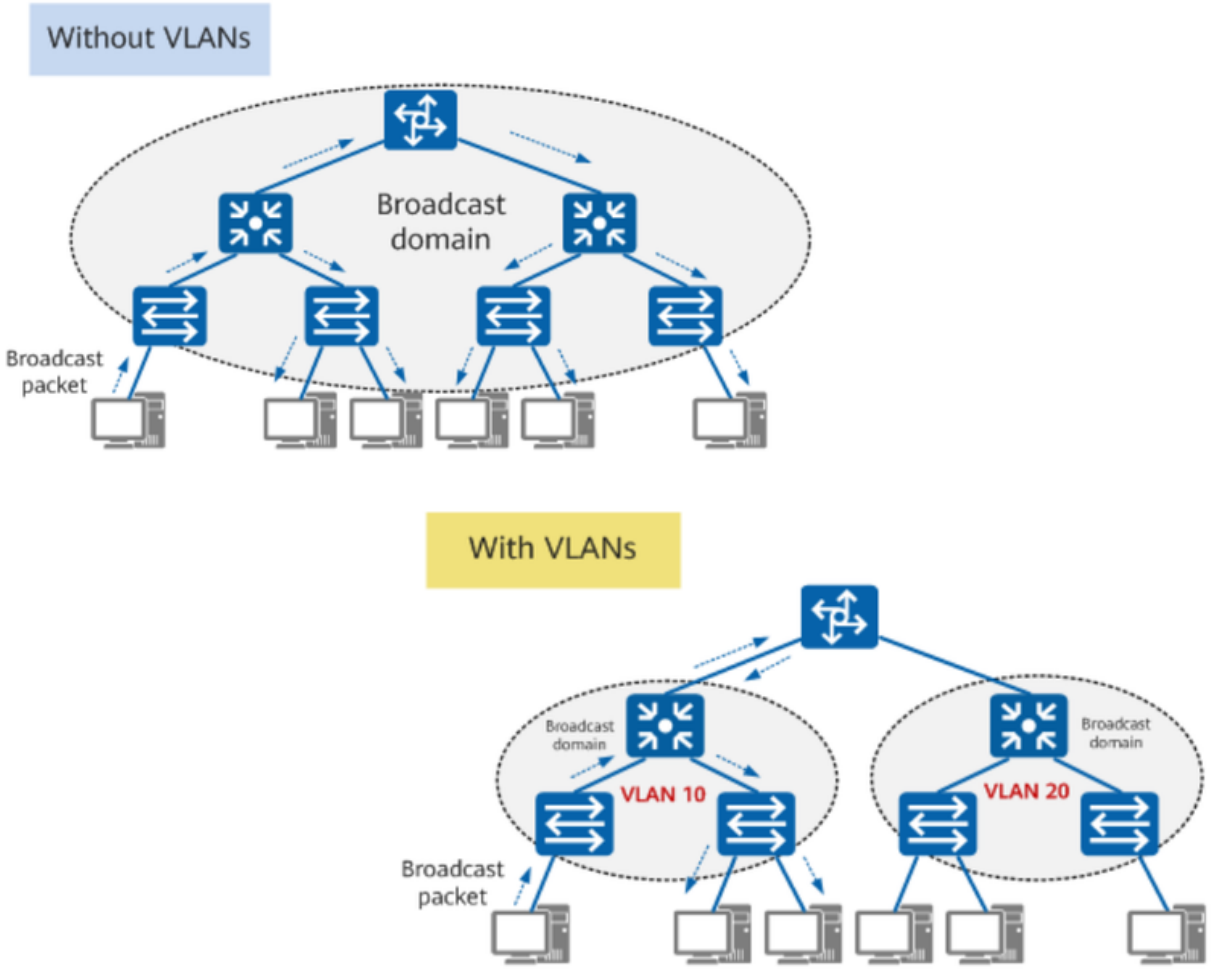


VLAN (Virtual Local Area Network) is a logical grouping of devices or users based on function, department, or application,.. where each VLAN is identified by a unique VLAN ID, a 12-bit value (ranging from 0 to 4095) defined in the IEEE 802.1Q standard.

- Same VLAN: Devices communicate directly within the same broadcast domain.
- Different VLANs: Communication requires Inter-VLAN Routing via a router or a Layer 3 switch

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Why we need VLAN?



- Smaller Broadcast Domains
- Improved Security
- Improved IT Efficiency
- Reduced Cost
- Better Performance
- Simpler Management

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Applications of VLAN

- Default VLAN (VLAN 1): Default VLAN for all ports; cannot be deleted or renamed
- Native VLAN: Used on trunk links; frames in this VLAN are sent untagged in 802.1Q
- Management VLAN: Used for remote management (SSH/Telnet); typically configured as the SVI on a Layer 2 switch and should be separated from user traffic
- Voice VLAN: Used for voice traffic; requires high QoS priority, guaranteed bandwidth, and low delay (<150 ms)
- Data VLAN: Used for user-generated traffic such as web and email

```
Switch# show vlan brief
VLAN Name                Status    Ports
---- ----------------- -------  -------------------
1    default             active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                   Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                   Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                   Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                   Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                   Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                   Gi0/1, Gi0/2
1002 fddi-default                        act/unsup
1003 token-ring-default                  act/unsup
1004 fddinet-default                     act/unsup
1005 trnet-default                       act/unsup
```



802.1Q tagged frame

The 802.1Q VLAN tag is inserted into a standard Ethernet frame immediately after the source MAC address. This tag consists of two main components:

1. TPID (Tag Protocol Identifier) – 2 bytes

A 16-bit field set to a value of 0x8100, indicating that the frame is an IEEE 802.1Q-tagged frame.

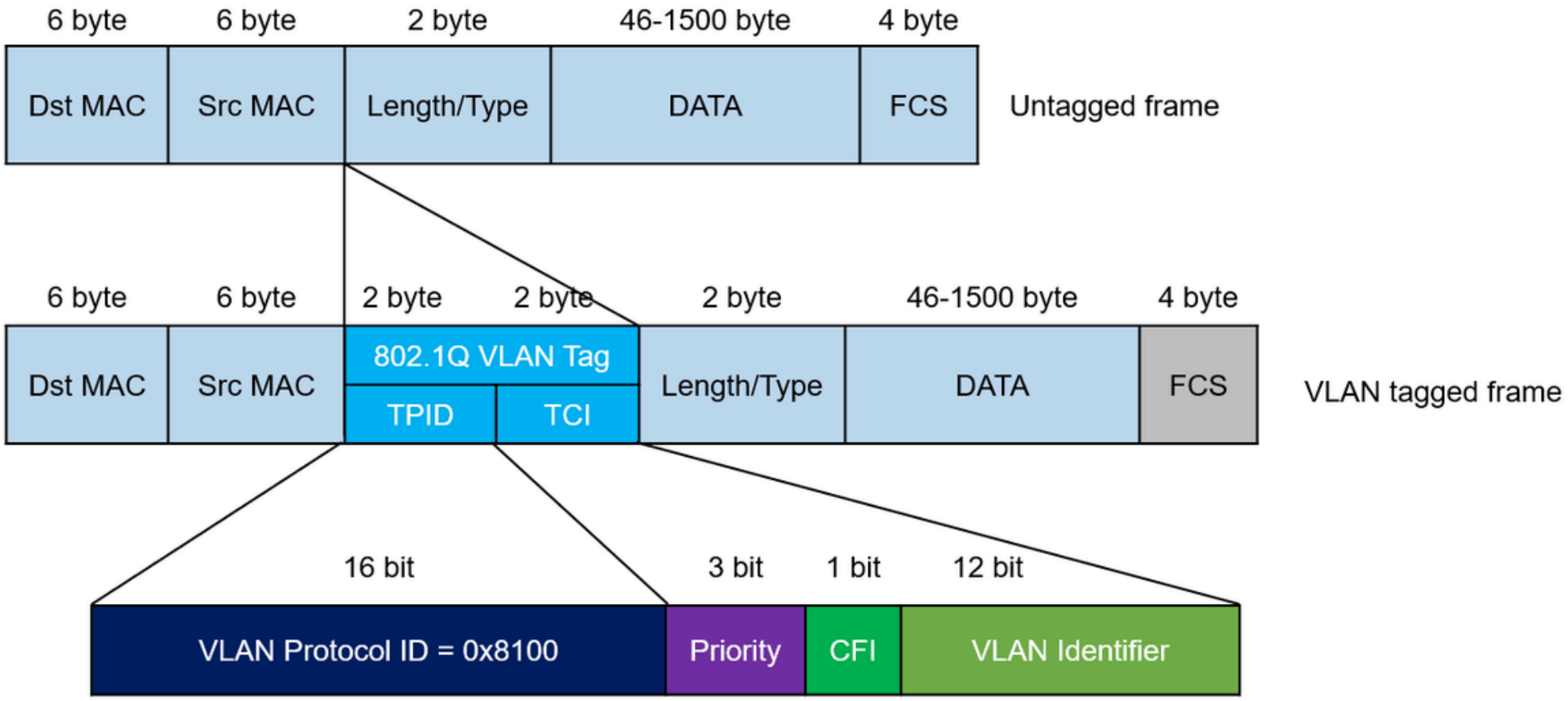
2. TCI (Tag Control Information)TCI– 2 bytes

This field contains the core control information and is divided into three subfields:

- Priority: A 3-bit field used for IEEE 802.1p priority, which indicates the frame's priority level for traffic prioritization.
- (CFI) Canonical Format Indicator: 1-bit field that specifies the format of the MAC address. A value of 0 indicates canonical format, while a value of 1 indicates non-canonical format.
- (VID) VLAN Identifier: A 12-bit field that uniquely identifies the VLAN to which the frame belongs. This field can have values ranging from 0 to 4095.

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802.1Q tagged frame

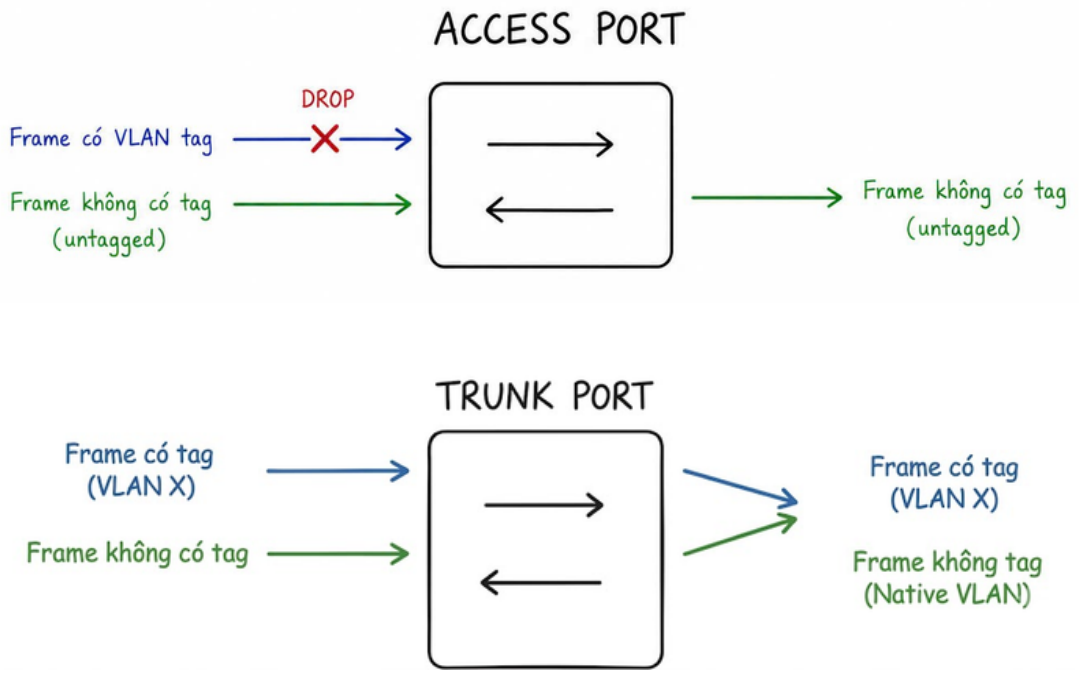


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VLAN Port Modes

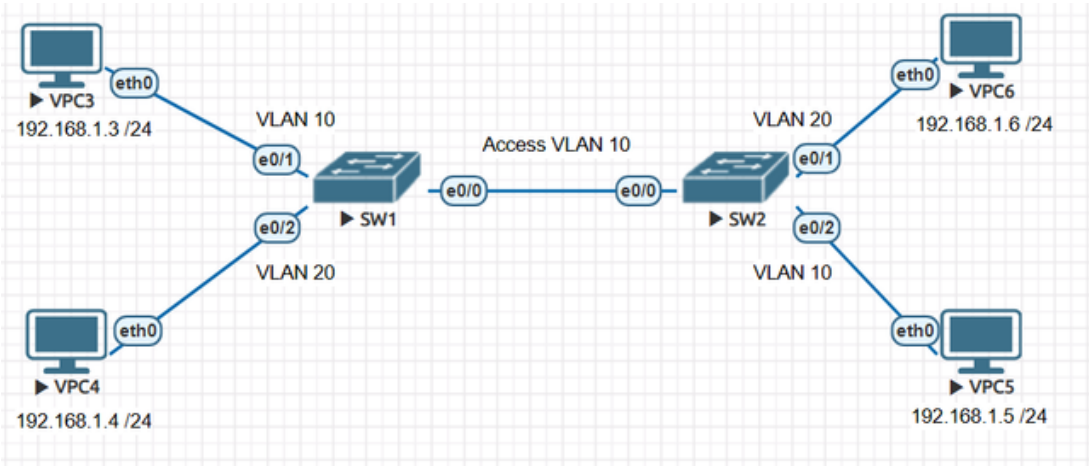
There are two main types of ports in VLANs:

- Access Port : Connects a VLAN-unaware end device (such as a PC) to a VLAN-aware switch. Frames transmitted over an access link are untagged and belong to a single VLAN.
- Trunk Port : Connects VLAN-aware devices such as switch-to-switch or switch-to-router links. It carries traffic from multiple VLANs using IEEE 802.1Q tagging, except for the native VLAN, which is transmitted untagged.



VLAN Port Modes

Access Port Verification & Packet Capture



10	13.546092	192.168.1.3	192.168.1.5	ICMP	98 Echo (ping) request id=0x5950
11	13.546799	192.168.1.5	192.168.1.3	ICMP	98 Echo (ping) reply id=0x5950
13	14.548865	192.168.1.3	192.168.1.5	ICMP	98 Echo (ping) request id=0x5a50
14	14.549444	192.168.1.5	192.168.1.3	ICMP	98 Echo (ping) reply id=0x5a50
15	15.551453	192.168.1.3	192.168.1.5	ICMP	98 Echo (ping) request id=0x5b50
16	15.552215	192.168.1.5	192.168.1.3	ICMP	98 Echo (ping) reply id=0x5b50
18	16.554070	192.168.1.3	192.168.1.5	ICMP	98 Echo (ping) request id=0x5c50
19	16.555071	192.168.1.5	192.168.1.3	ICMP	98 Echo (ping) reply id=0x5c50
20	17.556666	192.168.1.3	192.168.1.5	ICMP	98 Echo (ping) request id=0x5d50
21	17.557356	192.168.1.5	192.168.1.3	ICMP	98 Echo (ping) reply id=0x5d50

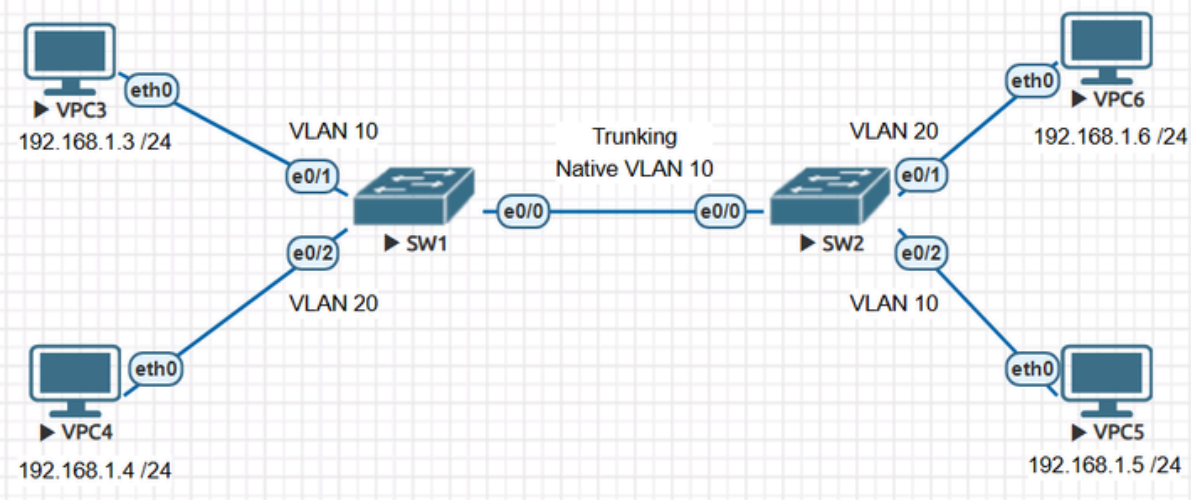
```
> Frame 10: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
v Ethernet II, Src: Private_66:68:03 (00:50:79:66:68:03), Dst: Private_66:68:05 (00:50:79:66:68:05)
  > Destination: Private_66:68:05 (00:50:79:66:68:05)
  > Source: Private_66:68:03 (00:50:79:66:68:03)
    Type: IPv4 (0x0800)
    [Stream index: 2]
> Internet Protocol Version 4, Src: 192.168.1.3, Dst: 192.168.1.5
> Internet Control Message Protocol
```

Traffic transmitted over the inter-switch access port consists of standard, untagged Ethernet frames (no 802.1Q header)

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VLAN Port Modes

Trunk Port Verification & Non-Native VLAN Packet Capture (Tagged)



18	6.853765	192.168.1.4	192.168.1.6	ICMP	102 Echo (ping) request
19	6.854474	192.168.1.6	192.168.1.4	ICMP	102 Echo (ping) reply
20	7.856626	192.168.1.4	192.168.1.6	ICMP	102 Echo (ping) request
21	7.857660	192.168.1.6	192.168.1.4	ICMP	102 Echo (ping) reply
26	8.859785	192.168.1.4	192.168.1.6	ICMP	102 Echo (ping) request
27	8.860497	192.168.1.6	192.168.1.4	ICMP	102 Echo (ping) reply
28	9.863176	192.168.1.4	192.168.1.6	ICMP	102 Echo (ping) request
29	9.863975	192.168.1.6	192.168.1.4	ICMP	102 Echo (ping) reply
34	10.865995	192.168.1.4	192.168.1.6	ICMP	102 Echo (ping) request
35	10.866762	192.168.1.6	192.168.1.4	ICMP	102 Echo (ping) reply

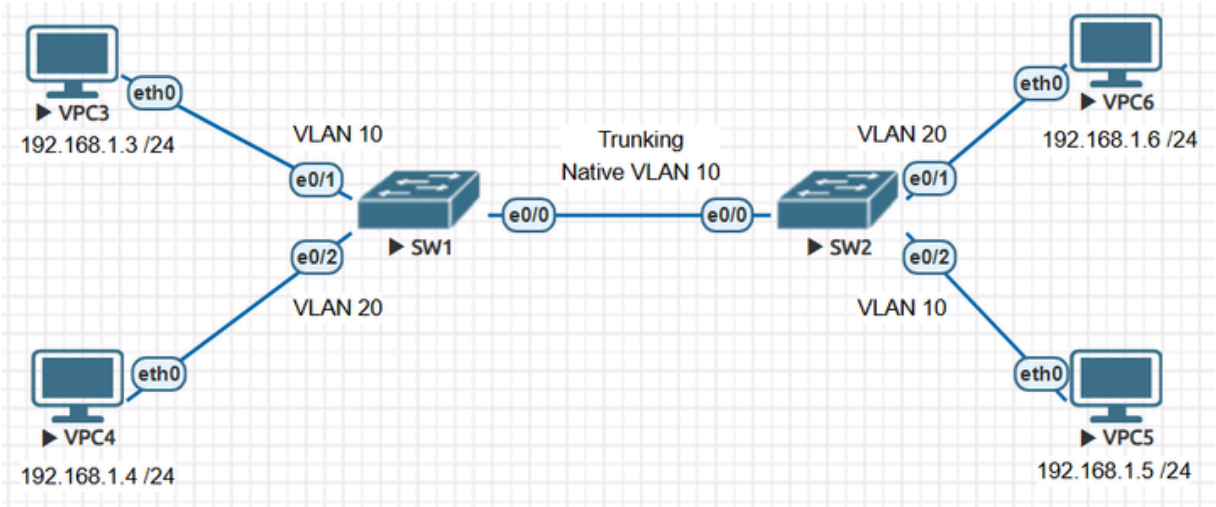
```
> Frame 18: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:04 (00:50:79:66:68:04), Dst: Private_66:68:06 (00:50:79:66:68:06)
v 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... .... .... = Priority: Best Effort (default) (0)
    ...0 .... .... .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: IPv4 (0x0800)
v Internet Protocol Version 4, Src: 192.168.1.4, Dst: 192.168.1.6
```

Frames belonging to non-native VLANs transmitted across a trunk link are tagged with an 802.1Q header to preserve VLAN identification between switches.

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VLAN Port Modes

Trunk Port Verification & Native VLAN Packet Capture (Untagged)




```
1545 728.528612 192.168.1.5 192.168.1.3 ICMP 98 Echo (ping) reply    id=0x
1550 729.529874 192.168.1.3 192.168.1.5 ICMP 98 Echo (ping) request  id=0x
1551 729.530756 192.168.1.5 192.168.1.3 ICMP 98 Echo (ping) reply    id=0x
1552 730.532624 192.168.1.3 192.168.1.5 ICMP 98 Echo (ping) request  id=0x
1553 730.533748 192.168.1.5 192.168.1.3 ICMP 98 Echo (ping) reply    id=0x
1558 731.535385 192.168.1.3 192.168.1.5 ICMP 98 Echo (ping) request  id=0x
1559 731.536099 192.168.1.5 192.168.1.3 ICMP 98 Echo (ping) reply    id=0x
1560 732.538385 192.168.1.3 192.168.1.5 ICMP 98 Echo (ping) request  id=0x
1561 732.539006 192.168.1.5 192.168.1.3 ICMP 98 Echo (ping) reply    id=0x

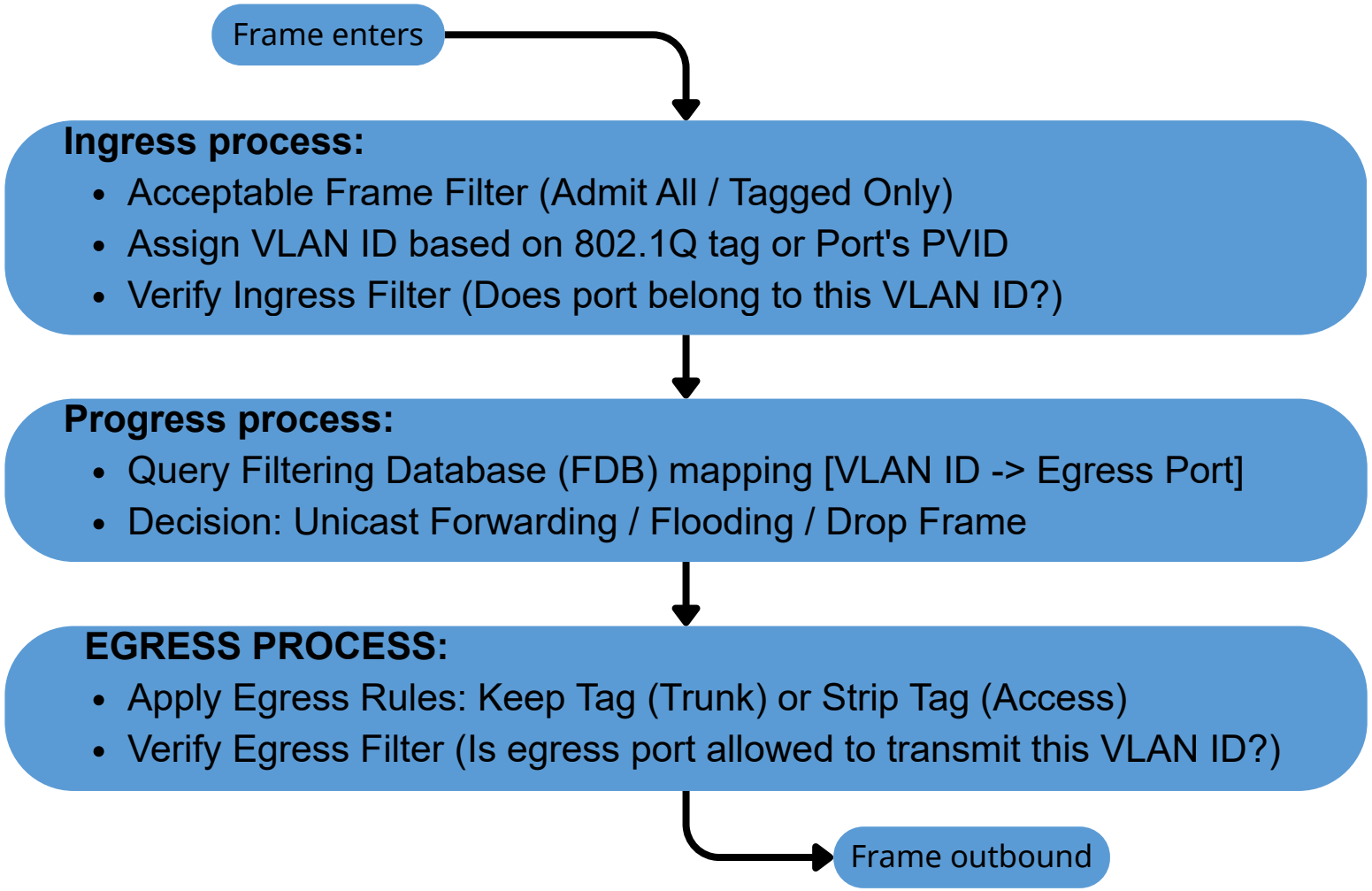
> Frame 1545: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
v Ethernet II, Src: Private_66:68:05 (00:50:79:66:68:05), Dst: Private_66:68:03 (00:50:79:66:68:03)
  > Destination: Private_66:68:03 (00:50:79:66:68:03)
  > Source: Private_66:68:05 (00:50:79:66:68:05)
    Type: IPv4 (0x0800)
    [Stream index: 9]
> Internet Protocol Version 4, Src: 192.168.1.5, Dst: 192.168.1.3
```

Traffic belonging to the Native VLAN transmitted across the trunk link is sent as standard, untagged Ethernet frames (no 802.1Q tag).

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VLAN FRAME HANDLING (Overview)

Ingress (Inbound) → Progress (Switching) → Egress (Outbound)



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VLAN FRAME HANDLING (Ingress processing)

Untagged

VLAN 20

Acceptable Frame Filter:

Configured port parameter to filter traffic:

- Admit All
- Admit All Tagged

Admit all

Untagged frame

VLAN 20

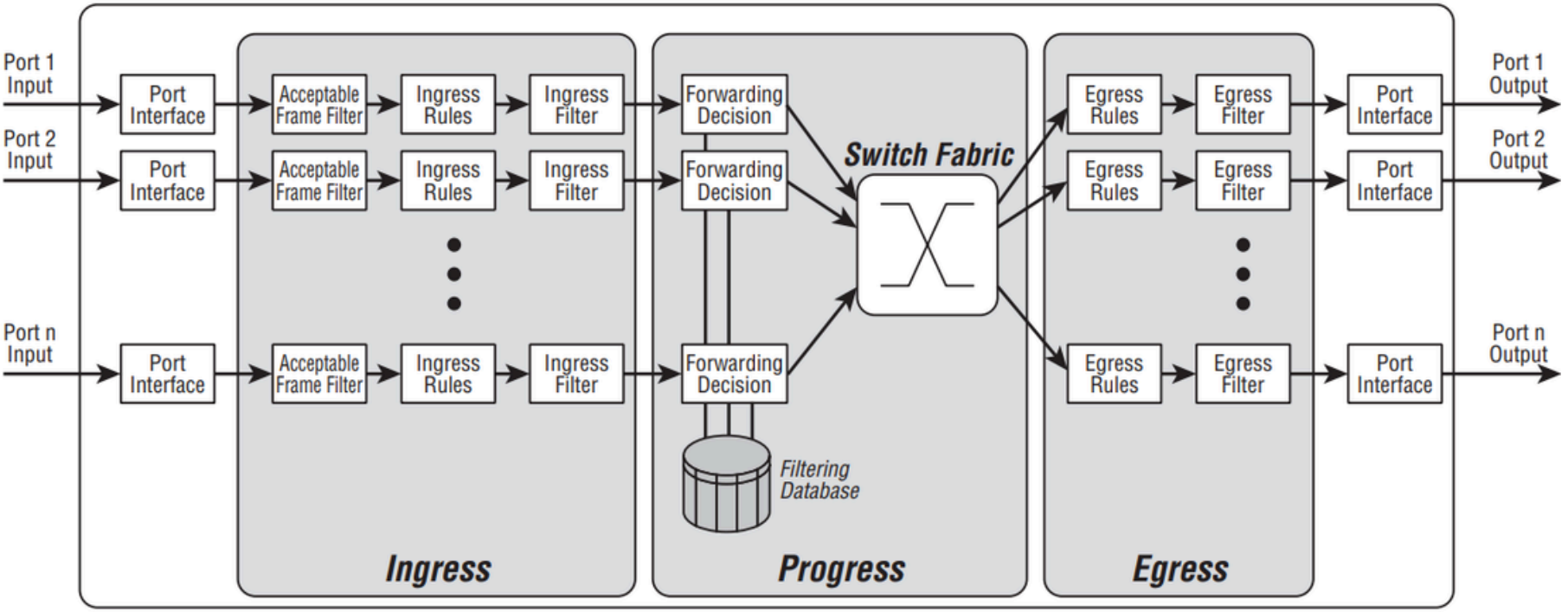
Ingress rules:

- If the received frame is untagged, asigning VLAN based on 802.1Q tag. (Association rules)
- If the received frame is Tagged, the switch preserves the VID value of the tag

PVID = 10

VLAN 20

VLAN 10



Ingress filtering:

- Check if the ingress port belongs to the Member Set of the assigned VLAN or not.

Drop (VLAN 20)

VLAN 10

Progress processing

VLAN FRAME HANDLING (Progress processing)

Tagged VLAN 10

Filltering Database:

Look up CAM table (MAC address table)

- Learning MAC
- Filtering
- Forwarding

Tagged VLAN 10

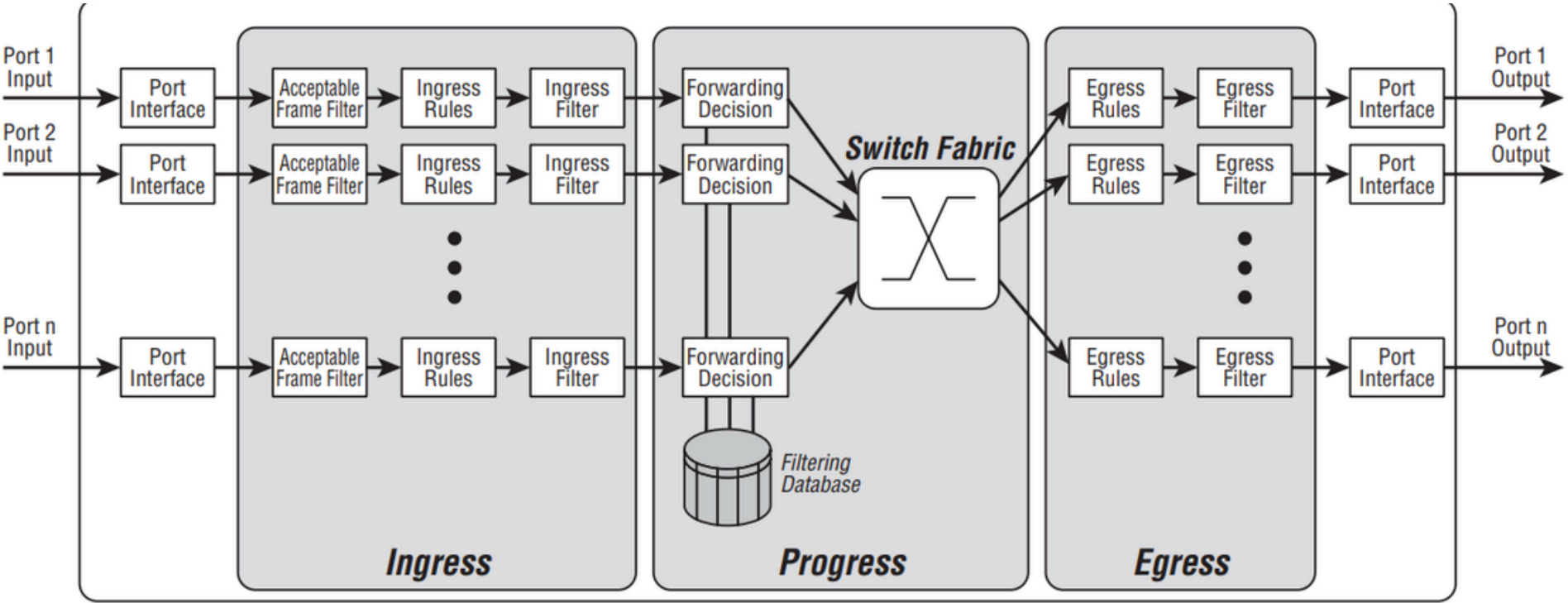
Forwarding Decision:

- Unicast forwarding
- Broadcast/ Unknown Unicast flooding

Tagged VLAN 10

Switch Fabric

Ergress processing



VLAN FRAME HANDLING (Egress processing)

Tagged VLAN 10

Egress Rules:

- Tagged/ Untagged frame decision

Port mode Access

untagged

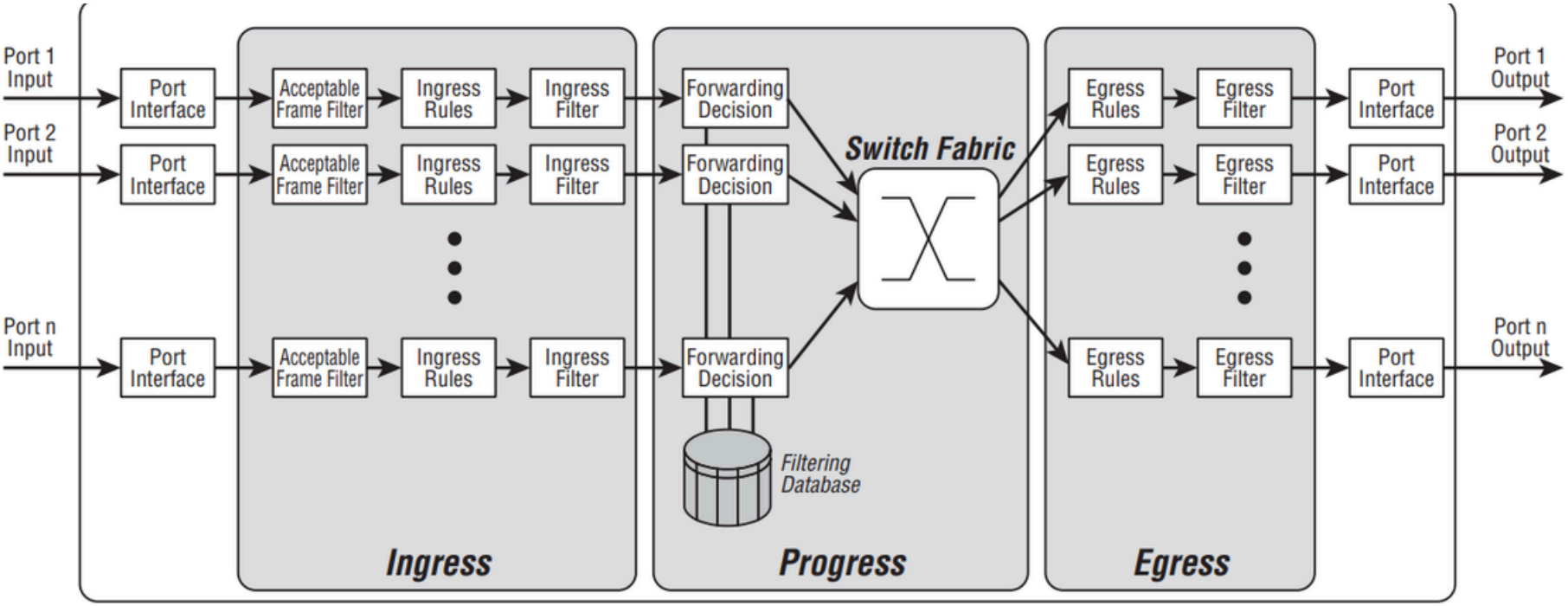
Egress Filter:

- Check if the ingress port belongs to the Member Set of the assigned VLAN or not.

PVID = 10

untagged

Port interface



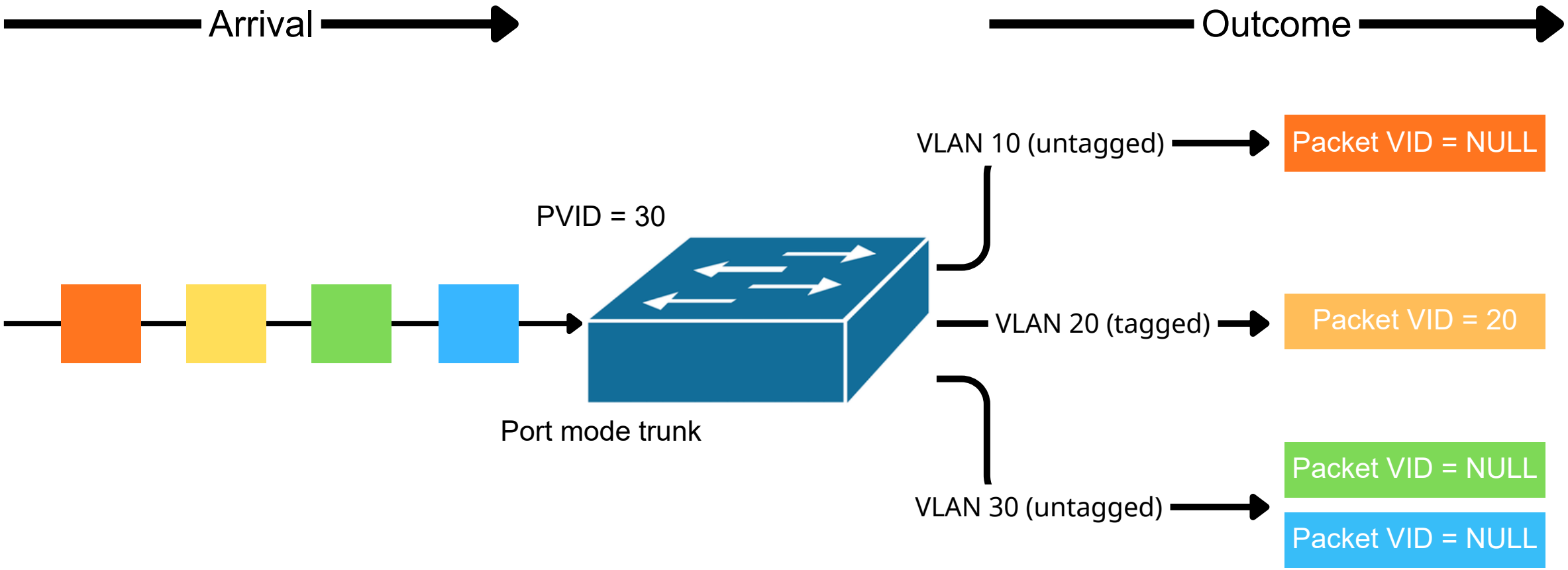
VLAN ASSOCIATION RULES (Ingress Rules)

Four types:

- **MAC-based:** Source MAC address
- **Subnet-based:** Source IP
- **Protocol-based:** Protocol use in upper layer
- **Port-based (default):** Port VID

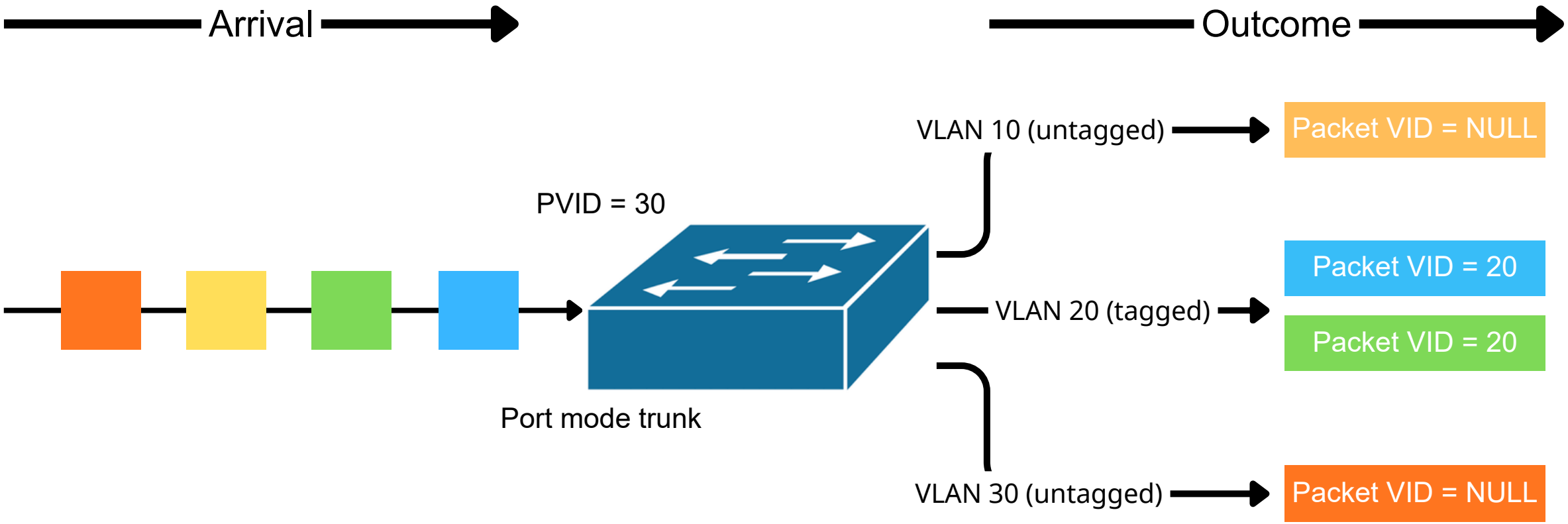
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PORT-BASED VLAN MECHANISM (Ingress Rules)



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MAC-BASED VLAN MECHANISM (Ingress Rules)



VLAN rules:
0:0:0:0:0:2 VLAN 20

0:0:0:0:0:1 VID = 30

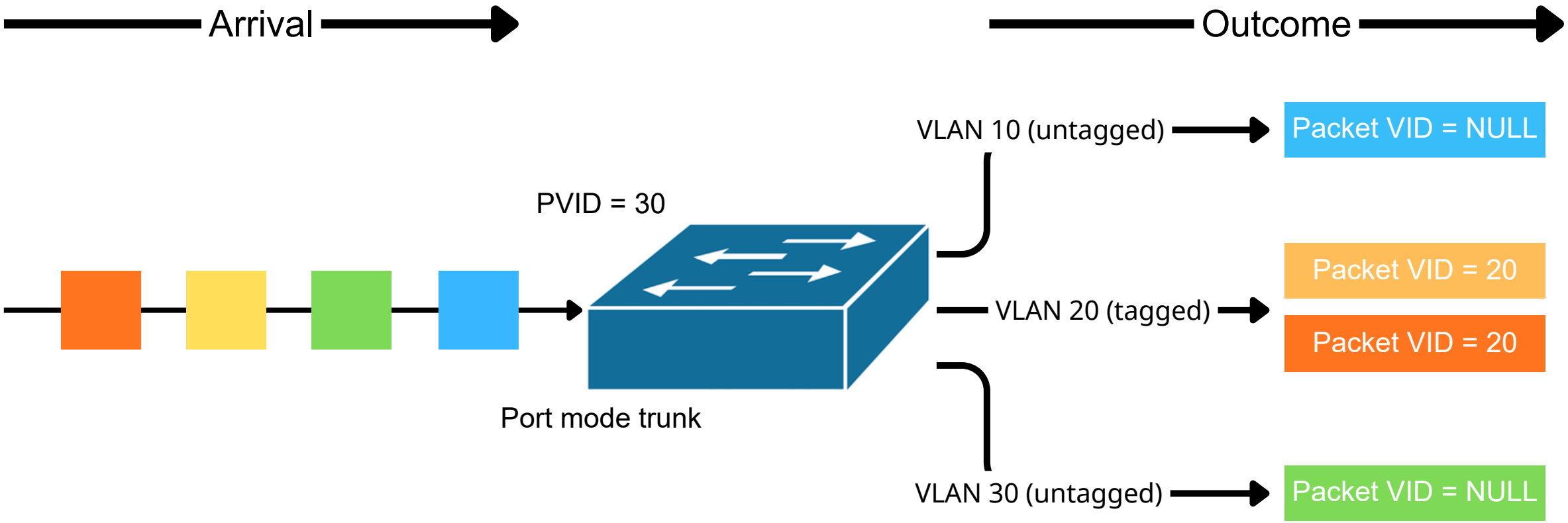
0:0:0:0:0:3 VID = 10

0:0:0:0:0:2 VID = NULL

0:0:0:0:0:4 VID = 20

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SUBNET-BASED VLAN MECHANISM (Ingress Rules)



VLAN rules:

10.0.0.1 VLAN 20

10.0.0.1 VID = NULL

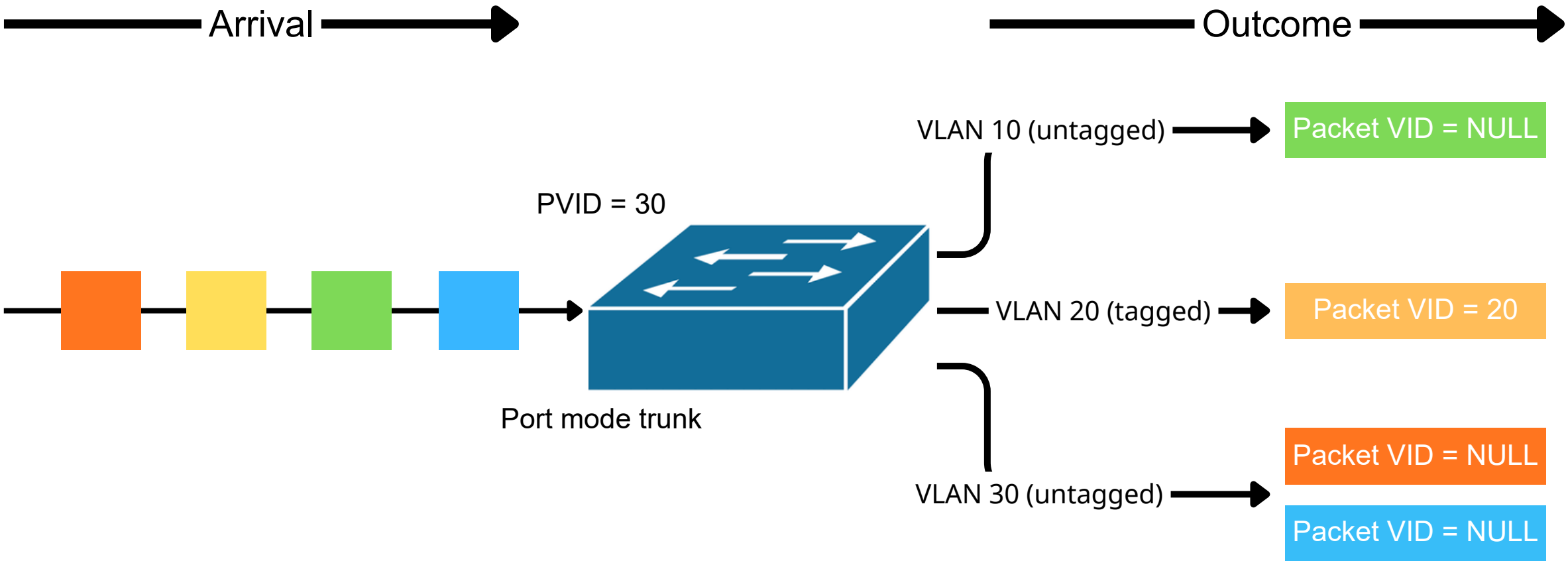
10.0.0.2 VID = 30

10.0.0.3 VID = 20

10.0.0.4 VID = 10

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PROTOCOL-BASED VLAN MECHANISM (Ingress Rules)



VLAN rules:
0x0806 VLAN 30

0x0800 VID = NULL

0x0802 VID = 20

0x0801 VID = 10

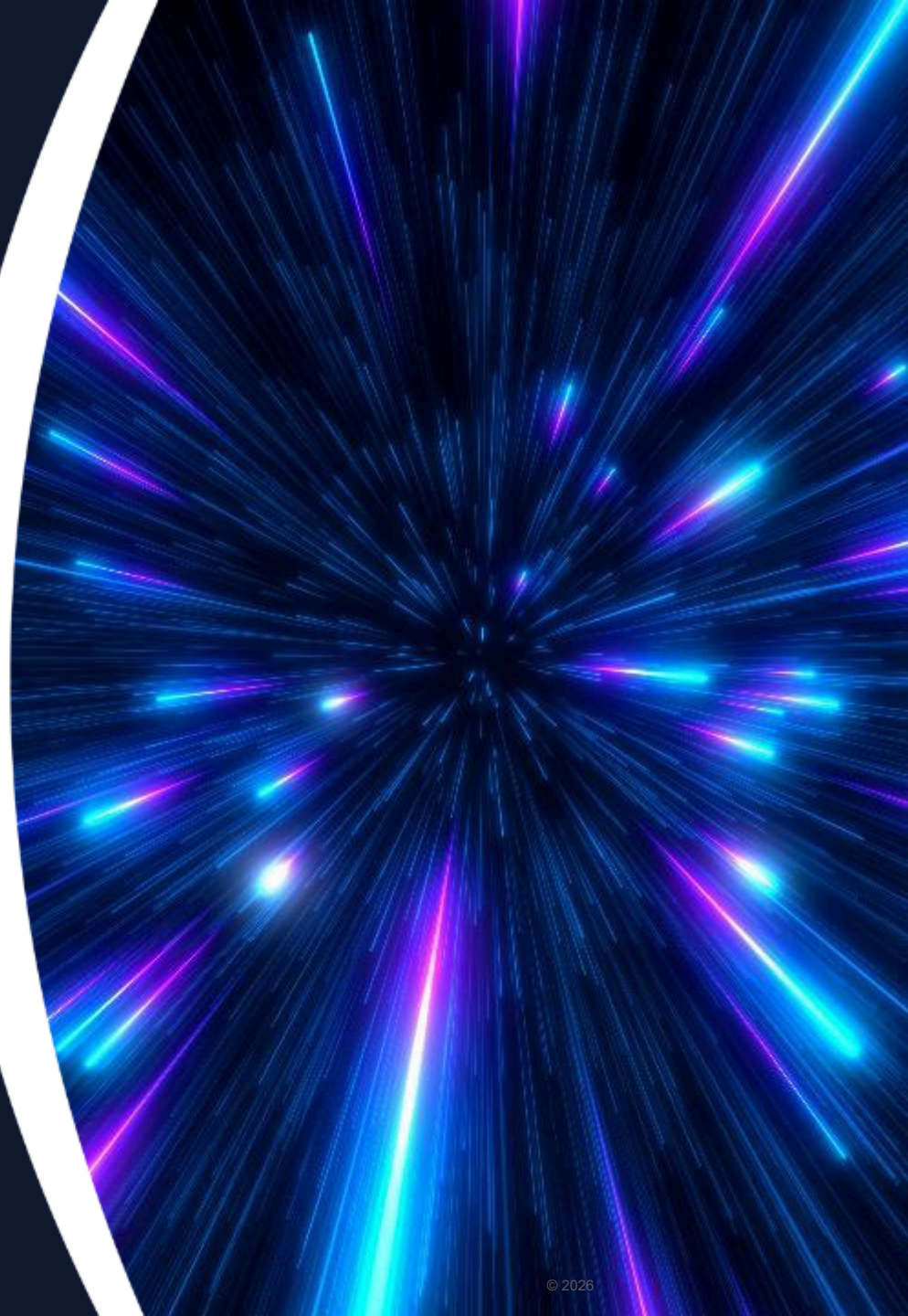
0x0806 VID = NULL

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Thank You For Listening!

Network QA Trainees



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